MATHCLEAR · VTU 2021 SCHEME

21MAT21 Complete Revision Handbook

Advanced Calculus and Numerical Methods

All module questions, full solutions, formula explanations, mark allocation and plain-English intuition.

60+ raw-mark preparation target

Preparation support—not an examination guarantee.

Preparation target: 60+ raw marks. Study above the bare minimum. No question set can guarantee a result.
Reliable method: Read the intuition → cover the solution → recall the formula → write the answer → compare and correct.
Source policy: Official VTU 2021 Scheme syllabus and model papers are the primary authority. Supplied VTU papers provide recurrence evidence. TIE/SIMP and other internet material are secondary cross-checks only where topics match the official syllabus.
VTU answer order: Given/definition → formula/theorem → substitution → stepwise working → boxed final answer.

Essential Full Forms

VTU: Visvesvaraya Technological University
SEE: Semester End Examination
CIE: Continuous Internal Evaluation
ODE: Ordinary Differential Equation
PDE: Partial Differential Equation
CF: Complementary Function
PI: Particular Integral
RK4: Fourth-Order Runge–Kutta Method
LHS / RHS: Left-Hand Side / Right-Hand Side

21MAT21 â€” Independently Authored VTU 2021-Scheme Practice Paper

Course: Advanced Calculus and Numerical Methods
Time: 3 Hours
Maximum Marks: 100

Instructions

Answer **five full questions**, choosing **one full question from each module**.
Each full question carries **20 marks**.
The observed VTU distribution of **6 + 7 + 7 marks** is followed.
Numerical answers are rounded only at the final stage unless an iteration requires displayed approximations.
The questions are independently written but selected from high-recurrence patterns in the supplied papers and restricted to the official syllabus.

Module 1 â€” Integral Calculus

Q1

Q1(a) Evaluate

$$I = \int_0^1 \int_0^2 (x + 2y), dy, dx. \qquad (6 \text{ marks})$$

Intuition: Integrate with respect to y first, treating x as a constant, and then integrate the resulting expression with respect to x.

Solution and marking scheme

Write the iterated integral correctly:

$$I = \int_0^1 \left[\int_0^2 (x + 2y), dy \right] dx.$$

[1 mark]

Integrate with respect to y:

$$\int (x + 2y), dy = xy + y^2.$$

[1 mark]

Apply the inner limits:

$$[xy + y^2]_0^2 = 2x + 4.$$

[1 mark]

Hence,

$$I = \int_0^1 (2x + 4), dx.$$

[1 mark]

Integrate:

$$I = [x^2 + 4x]_0^1.$$

[1 mark]

Therefore,

$$\boxed{I = 5}.$$

[1 mark]

Q1(b) Change the order of integration and evaluate

$$I = \int_0^1 \int_x^1 (x+y), dy, dx. \qquad (7 \text{ marks})$$

Intuition: The original region satisfies $0 \le x \le 1$ and $x \le y \le 1$. Horizontally, the same triangle is described by $0 \le$ ≤ 1 and $0 \le x \le y$.

Solution and marking scheme

Identify the original region:

$$0 \le x \le 1, \qquad x \le y \le 1.$$

[1 mark]

Its boundaries are $x = 0$, $y = x$, and $y = 1$. After reversing the order:

$$0 \le y \le 1, \qquad 0 \le x \le y.$$

[2 marks]

Rewrite the integral:

$$I = \int_0^1 \int_0^y (x+y), dx, dy.$$

[1 mark]

Integrate with respect to x:

$$\int_0^y (x+y), dx = \left[\frac{x^2}{2} + yx\right]_0^y.$$

[1 mark]

Simplify:

$$\frac{y^2}{2} + y^2 = \frac{3y^2}{2}.$$

[1 mark]

Therefore,

$$I = \frac{3}{2}\int_0^1 y^2, dy = \frac{3}{2}\left[\frac{y^3}{3}\right]_0^1 = \boxed{\frac{1}{2}}.$$

[1 mark]

Q1© Derive the relation between the Beta and Gamma functions.

(7 marks)

Intuition: Multiply two Gamma integrals, transform the first quadrant using $x = ru$ and $y = r(1-u)$, and separate the result into a Gamma integral and a Beta integral.

Solution and marking scheme

Definitions:

$$\Gamma(m) = \int_0^\infty e^{-x} x^{m-1}, dx, \qquad B(m,n) = \int_0^1 u^{m-1}(1-u)^{n-1}, du.$$

[1 mark]

Multiply the two Gamma functions:

$$\Gamma(m)\Gamma(n) = \int_0^\infty \int_0^\infty e^{-(x+y)} x^{m-1} y^{n-1}, dx, dy.$$

[1 mark]

Put

$$x = ru, \qquad y = r(1-u),$$

where $r \in [0, \infty)$ and $u \in [0, 1]$. **[1 mark]**

The Jacobian is

$$\left|\frac{\partial(x,y)}{\partial(r,u)}\right| = r.$$

[1 mark]

Therefore,

$$\Gamma(m)\Gamma(n) = \int_0^\infty \int_0^1 e^{-r} r^{m+n-1} u^{m-1}(1-u)^{n-1}, du, dr.$$

[1 mark]

Separate the integrals: $$ \Gamma(m)\Gamma(n)

\left[\int_0^\infty e^{-r}r^{m+n-1},dr\right] \left[\int_0^1u^{m-1}(1-u)^{n-1},du\right].

$$Thus,$$

\Gamma(m)\Gamma(n)=\Gamma(m+n)B(m,n). $$ **[1 mark]**
Hence,

$$\boxed{B(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}}.$$

[1 mark]

OR

Q2

Q2(a) Convert into polar coordinates and evaluate

$$I = \iint_R (x^2 + y^2), dA,$$

where R is the part of the unit circle in the first quadrant.

(6 marks)

Intuition: In polar coordinates, $x^2 + y^2 = r^2$ and the area element is $dA = r, dr, d\theta$.

Solution and marking scheme

For the first-quadrant unit disk:

$$0 \le r \le 1, \qquad 0 \le \theta \le \frac{\pi}{2}.$$

[1 mark]

Use

$$x = r\cos\theta, \quad y = r\sin\theta, \quad dA = r, dr, d\theta.$$

[1 mark]

Since $x^2 + y^2 = r^2$,

$$I = \int_0^{\pi/2} \int_0^1 r^3, dr, d\theta.$$

[1 mark]

Evaluate the radial integral:

$$\int_0^1 r^3, dr = \frac{1}{4}.$$

[1 mark]

Thus,

$$I = \frac{1}{4} \int_0^{\pi/2} d\theta.$$

[1 mark]

Therefore,

$$\boxed{I = \frac{\pi}{8}}.$$

[1 mark]

Q2(b) Using double integration, find the area enclosed by

$$y = x^2, \qquad x = y^2$$

in the first quadrant.

(7 marks)

Intuition: In the first quadrant, $x = y^2$ gives $y = \sqrt{x}$. Between the intersection points, $\sqrt{x}$ is above x^2.

Solution and marking scheme

Solve the curves simultaneously:

$$y = x^2, \qquad x = y^2.$$

Substituting $y = x^2$:

$$x = x^4.$$

[1 mark]

Hence,

$$x(x^3 - 1) = 0 \implies x = 0, 1.$$

The intersection points are $(0, 0)$ and $(1, 1)$. **[1 mark]**

Rewrite $x = y^2$ as

$$y = \sqrt{x}$$

in the first quadrant. **[1 mark]**

The required region is

$$0 \le x \le 1, \qquad x^2 \le y \le \sqrt{x}.$$

[1 mark]

Therefore,

$$A = \int_0^1 \int_{x^2}^{\sqrt{x}} dy, dx = \int_0^1 (\sqrt{x} - x^2), dx.$$

[1 mark]

Integrate:

$$A = \left[\frac{2}{3}x^{3/2} - \frac{x^3}{3}\right]_0^1.$$

[1 mark]

Hence,

$$\boxed{A = \frac{2}{3} - \frac{1}{3} = \frac{1}{3} \text{ square unit}}.$$

[1 mark]

Q2© Using Beta and Gamma functions, evaluate

$$I = \int_0^{\pi/2} \sin^3\theta \cos^2\theta, d\theta. \qquad (7 \text{ marks})$$

Intuition: Match the powers with the standard trigonometric Beta integral.

Solution and marking scheme

Use

$$\int_0^{\pi/2} \sin^{m-1}\theta \cos^{n-1}\theta, d\theta = \frac{1}{2} B\left(\frac{m}{2}, \frac{n}{2}\right).$$

[1 mark]

Here,

$$m-1=3, \qquad n-1=2,$$

so $m=4$ and $n=3$. **[1 mark]**

Thus,

$$I = \frac{1}{2} B\left(2, \frac{3}{2}\right).$$

[1 mark]

Apply the Beta-Gamma relation:

$$I = \frac{1}{2} \frac{\Gamma(2)\Gamma(3/2)}{\Gamma(7/2)}.$$

[1 mark]

Evaluate:

$$\Gamma(2) = 1, \qquad \Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}.$$

[1 mark]

Also,

$$\Gamma\left(\frac{7}{2}\right) = \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi} = \frac{15\sqrt{\pi}}{8}.$$

[1 mark]

Therefore,

$$I = \frac{1}{2} \frac{\sqrt{\pi}/2}{15\sqrt{\pi}/8} = \boxed{\frac{2}{15}}.$$

[1 mark]

Module 1 formula glossary

Beta function:

$$B(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}, dx.$$

Gamma function:

$$\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} , dx.$$

$\Gamma(n+1) = n\Gamma(n)$ and $\Gamma(1/2) = \sqrt{\pi}$.
$B(m,n) = \dfrac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$.

Polar transformation: $x = r\cos\theta, y = r\sin\theta, dA = r, dr, d\theta$.

Jacobian: Scale factor used when variables in a multiple integral are changed.

Module 2 â€” Vector Calculus

Q3

Q3(a) Find the directional derivative of

$$\phi = x^2y + yz^2$$

at $(1, 2, -1)$ in the direction $2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$.

(6 marks)

Intuition: The directional derivative is the dot product of the gradient with the unit direction vector.

Solution and marking scheme

Compute the gradient:

$$\nabla\phi = \left(2xy,\ x^2 + z^2,\ 2yz\right).$$

[1 mark]

At $(1, 2, -1)$:

$$\nabla\phi = (4, 2, -4).$$

[1 mark]

The given direction vector is

$$\mathbf{a} = (2, -1, 2).$$

[1 mark]

Its magnitude is

$$|\mathbf{a}| = \sqrt{4 + 1 + 4} = 3.$$

Therefore,

$$\hat{\mathbf{a}} = \frac{1}{3}(2, -1, 2).$$

[1 mark]

The directional derivative is

$$D_{\hat{\mathbf{a}}}\phi = \nabla\phi \cdot \hat{\mathbf{a}} = \frac{1}{3}(8-2-8).$$

[1 mark]

Hence,

$$\boxed{D_{\hat{\mathbf{a}}}\phi = -\frac{2}{3}}.$$

[1 mark]

Q3(b) For

$$\mathbf{F} = xy, \mathbf{i} + yz, \mathbf{j} + zx, \mathbf{k},$$

find $\operatorname{div}\mathbf{F}$ and $\operatorname{curl}\mathbf{F}$. Also evaluate them at $(1,2,3)$.

(7 marks)

Intuition: Divergence measures net outward spreading; curl measures local rotation.

Solution and marking scheme

Let

$$P = xy, \qquad Q = yz, \qquad R = zx.$$

[1 mark]

Divergence:

$$\nabla \cdot \mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}.$$

[1 mark]

Therefore,

$$\nabla \cdot \mathbf{F} = y + z + x.$$

At $(1,2,3)$:

$$\boxed{\nabla \cdot \mathbf{F} = 6}.$$

[1 mark]

Curl:

$$\nabla \times \mathbf{F} = |\mathbf{i} \quad \mathbf{j} \quad \mathbf{k}\, \partial_x \quad \partial_y \quad \partial_z\, xy \quad yz \quad zx|\,.$$

[1 mark]

Its $\mathbf{i}$-component is

$$\frac{\partial(zx)}{\partial y} - \frac{\partial(yz)}{\partial z} = 0 - y = -y.$$

[1 mark]

Similarly,

$$\mathbf{j}\text{-component} = -z, \qquad \mathbf{k}\text{-component} = -x.$$

Hence,

$$\nabla \times \mathbf{F} = -y\mathbf{i} - z\mathbf{j} - x\mathbf{k}.$$

[1 mark]

At $(1, 2, 3)$:

$$\boxed{\nabla \times \mathbf{F} = -2\mathbf{i} - 3\mathbf{j} - \mathbf{k}}.$$

[1 mark]

Q3© Find a, b, c so that

$$\mathbf{F} = (ay + z)\mathbf{i} + (2x + bz)\mathbf{j} + (cy + cx)\mathbf{k}$$

is irrotational.

(7 marks)

Intuition: An irrotational field has zero curl, so each component of $\nabla \times \mathbf{F}$ must vanish.

Solution and marking scheme

Write

$$P = ay + z, \quad Q = 2x + bz, \quad R = cy + cx.$$

[1 mark]

The first curl component is

$$R_y - Q_z = c - b.$$

[1 mark]

The second component is

$$P_z - R_x = 1 - c.$$

[1 mark]

The third component is

$$Q_x - P_y = 2 - a.$$

[1 mark]

For an irrotational field:

$$c - b = 0, \qquad 1 - c = 0, \qquad 2 - a = 0.$$

[1 mark]

Hence,

$$c = 1, \qquad b = c = 1, \qquad a = 2.$$

[1 mark]

Therefore,

$$\boxed{a = 2, \quad b = 1, \quad c = 1}.$$

[1 mark]

OR

Q4

Q4(a) Find the work done by

$$\mathbf{F} = 2x, \mathbf{i} + y, \mathbf{j} + z, \mathbf{k}$$

along the straight line from $(0,0,0)$ to $(1,2,3)$.

(6 marks)

Intuition: Parametrize the line and evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$.

Solution and marking scheme

Parametrize:

$$\mathbf{r}(t) = (t, 2t, 3t), \qquad 0 \le t \le 1.$$

[1 mark]

Then

$$d\mathbf{r} = (1, 2, 3), dt.$$

[1 mark]

Along the curve:

$$\mathbf{F} = (2t, 2t, 3t).$$

[1 mark]

Thus,

$$\mathbf{F} \cdot d\mathbf{r} = (2t + 4t + 9t), dt = 15t, dt.$$

[1 mark]

Hence,

$$W = \int_0^1 15t, dt.$$

[1 mark]

Therefore,

$$\boxed{W = \frac{15}{2}}.$$

[1 mark]

Q4(b) Apply Greenâ€™s theorem to evaluate

$$\oint_C (x^2 - y), dx + (x + y^2), dy,$$

where C is the positively oriented boundary of the unit square.

(7 marks)

Intuition: Greenâ€™s theorem replaces the boundary integral by a double integral over the square.

Solution and marking scheme

Identify

$$M = x^2 - y, \qquad N = x + y^2.$$

[1 mark]

Greenâ€™s theorem:

$$\oint_C M, dx + N, dy = \iint_R (N_x - M_y)\, dA.$$

[1 mark]

Compute

$$N_x = 1.$$

[1 mark]

Compute

$$M_y = -1.$$

[1 mark]

Thus,

$$N_x - M_y = 1 - (-1) = 2.$$

[1 mark]

For $0 \le x, y \le 1$:

$$\iint_R 2, dA = 2 \int_0^1 \int_0^1 dy, dx.$$

[1 mark]

Therefore,

$$\boxed{\oint_C M, dx + N, dy = 2}.$$

[1 mark]

Q4© Using Stokesâ€™ theorem, evaluate

$$\oint_C (-y, dx + x, dy),$$

where C is the counterclockwise boundary of

$$0 \le x \le 2, \qquad 0 \le y \le 1$$

in the xy-plane.

(7 marks)

Intuition: Treat the integrand as $\mathbf{F} \cdot d\mathbf{r}$ and replace the line integral by the flux of $\nabla \times \mathbf{F}$.

Solution and marking scheme

Take

$$\mathbf{F} = -y, \mathbf{i} + x, \mathbf{j}.$$

[1 mark]

Stokesâ€™ theorem gives

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \widehat{\mathbf{n}}, dS.$$

[1 mark]

Compute: $$ \nabla\times\mathbf F

$$|\mathbf{i} \quad \mathbf{j} \quad \mathbf{k}\, \partial_x \quad \partial_y \quad \partial_z \, -y \quad x \quad 0|$$

. $$ **[1 mark]**

Therefore,

$$\nabla \times \mathbf{F} = 2\mathbf{k}.$$

[1 mark]

For counterclockwise orientation,

$$\widehat{\mathbf{n}} = \mathbf{k}.$$

Hence the integrand is 2. **[1 mark]**

The rectangle has area $2 \times 1 = 2$. **[1 mark]**

Consequently,

$$\boxed{\oint_C (-y, dx + x, dy) = 2(2) = 4}.$$

[1 mark]

Module 2 formula glossary

Gradient: $\nabla\phi$; points in the direction of maximum increase.

Directional derivative: $D_{\hat{\mathbf{a}}}\phi = \nabla\phi \cdot \hat{\mathbf{a}}$.

Divergence: $\operatorname{div}\mathbf{F} = \nabla \cdot \mathbf{F}$.

Curl: $\operatorname{curl}\mathbf{F} = \nabla \times \mathbf{F}$.

Solenoidal: $\nabla \cdot \mathbf{F} = 0$.

Irrotational: $\nabla \times \mathbf{F} = \mathbf{0}$.

Greenâ€™s theorem:

$$\oint_C M, dx + N, dy = \iint_R (N_x - M_y), dA.$$

Stokesâ€™ theorem:

$$\oint_C \mathbf{F}\cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \widehat{\mathbf{n}}, dS.$$

Module 3 â€” Partial Differential Equations

Q5

Q5 (a) Form the PDE by eliminating the arbitrary function from

$$z = f(x + 2y). \qquad (6 \text{ marks})$$

Intuition: Differentiate with respect to both independent variables and eliminate f'.

Solution and marking scheme

Let

$$u = x + 2y, \qquad z = f(u).$$

[1 mark]

Put

$$p = \frac{\partial z}{\partial x}, \qquad q = \frac{\partial z}{\partial y}.$$

[1 mark]

Differentiate with respect to x:

$$p = f'(u)\frac{\partial u}{\partial x} = f'(u).$$

[1 mark]

Differentiate with respect to y:

$$q = f'(u)\frac{\partial u}{\partial y} = 2f'(u).$$

[1 mark]

Since $p = f'(u)$,

$$q = 2p.$$

[1 mark]

Therefore, the required PDE is

$$\boxed{q - 2p = 0}$$

or

$$\boxed{\frac{\partial z}{\partial y} - 2\frac{\partial z}{\partial x} = 0}.$$

[1 mark]

Q5(b) Solve

$$\frac{\partial^2 z}{\partial x^2} = 6xy,$$

given

$$\frac{\partial z}{\partial x} = y \quad \text{at } x = 0, \qquad z = y^2 \quad \text{at } x = 0. \qquad\qquad (7 \text{ marks})$$

Intuition: Integrate twice with respect to x. The â€œconstantsâ€ of integration may be functions of y.

Solution and marking scheme

Integrate once with respect to x:

$$z_x = 3x^2y + A(y).$$

[1 mark]

At $x = 0$, $z_x = y$. Therefore,

$$A(y) = y.$$

[1 mark]

Hence,

$$z_x = 3x^2y + y.$$

[1 mark]

Integrate again:

$$z = x^3y + xy + B(y).$$

[1 mark]

At $x = 0$, $z = y^2$. Hence,

$$B(y) = y^2.$$

[1 mark]

Therefore,

$$z = x^3y + xy + y^2.$$

[1 mark]

Check:

$$z_{xx} = 6xy.$$

Thus,

$$\boxed{z = x^3y + xy + y^2}.$$

[1 mark]

Q5© Derive the one-dimensional heat equation.

(7 marks)

Intuition: Equate the net heat entering a small rod element to the rate of increase of thermal energy within it.

Solution and marking scheme

Consider a uniform rod with cross-sectional area A, density ρ, specific heat c, and temperature $u(x, t)$. **[1 mark]**

By Fourierâ€™s law, heat flow rate in the positive x-direction is

$$-KA\frac{\partial u}{\partial x},$$

where K is thermal conductivity. **[1 mark]**

Heat entering at x during time dt is

$$-KA, u_x(x, t), dt.$$

[1 mark]

Heat leaving at $x + dx$ is

$$-KA\left[u_x + \frac{\partial u_x}{\partial x}dx\right]dt.$$

[1 mark]

Therefore, net heat gained is

$$KA, u_{xx}, dx, dt.$$

[1 mark]

Increase in internal thermal energy is

$$\rho c A, dx, u_t, dt.$$

Equating heat gained and energy increase:

$$KA, u_{xx}, dx, dt = \rho c A, dx, u_t, dt.$$

[1 mark]

Therefore,

$$\boxed{\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}}, \qquad \boxed{\alpha^2 = \frac{K}{\rho c}}.$$

[1 mark]

OR

Q6

Q6 (a) Form the PDE by eliminating a and b from

$$z = (x-a)^2 + (y-b)^2. \qquad \text{(6 marks)}$$

Intuition: Differentiate once with respect to x and y, solve for the shifted coordinates, and substitute into the original equation.

Solution and marking scheme

Put

$$p = z_x, \qquad q = z_y.$$

[1 mark]

Differentiate with respect to x:

$$p = 2(x - a).$$

[1 mark]

Differentiate with respect to y:

$$q = 2(y - b).$$

[1 mark]

Hence,

$$x - a = \frac{p}{2}, \qquad y - b = \frac{q}{2}.$$

[1 mark]

Substitute into the original relation:

$$z = \frac{p^2}{4} + \frac{q^2}{4}.$$

[1 mark]

Therefore,

$$\boxed{p^2 + q^2 = 4z}.$$

[1 mark]

Q6 (b) Solve Lagrangeâ€™s PDE

$$(y - z)p + (z - x)q = x - y. \qquad \text{(7 marks)}$$

Intuition: Use the auxiliary equations and find two independent first integrals.

Solution and marking scheme

The auxiliary equations are

$$\frac{dx}{y - z} = \frac{dy}{z - x} = \frac{dz}{x - y}.$$

[1 mark]

Choose multipliers $(1, 1, 1)$. Since

$$(y - z) + (z - x) + (x - y) = 0,$$

we obtain

$$dx + dy + dz = 0.$$

[1 mark]

Integrating:

$$x + y + z = C_1.$$

[1 mark]

Choose multipliers (x, y, z). Then

$$x(y - z) + y(z - x) + z(x - y) = 0.$$

[1 mark]

Hence,

$$x, dx + y, dy + z, dz = 0.$$

[1 mark]

Integrating:

$$x^2 + y^2 + z^2 = C_2.$$

[1 mark]

Therefore, the complete solution is

$$\boxed{F\left(x + y + z,\ x^2 + y^2 + z^2\right) = 0},$$

where F is arbitrary. **[1 mark]**

Q6© Solve

$$\frac{\partial^2 z}{\partial y^2} = 4z,$$

given that at $y = 0$,

$$z = e^x, \qquad \frac{\partial z}{\partial y} = 0. \qquad \text{(7 marks)}$$

Intuition: Treat x as a parameter and solve the equation as an ordinary differential equation in y.

Solution and marking scheme

Write the equation as

$$z_{yy} - 4z = 0.$$

[1 mark]

The auxiliary equation is

$$m^2 - 4 = 0.$$

[1 mark]

Thus,

$$m = \pm 2.$$

[1 mark]

The general solution is

$$z = A(x)e^{2y} + B(x)e^{-2y}.$$

[1 mark]

At $y = 0$:

$$A(x) + B(x) = e^x.$$

[1 mark]

Since

$$z_y = 2A(x)e^{2y} - 2B(x)e^{-2y},$$

the second condition gives $A(x) = B(x) = e^x/2$. **[1 mark]**

Therefore,

$$z = \frac{e^x}{2}(e^{2y} + e^{-2y}) = \boxed{e^x \cosh 2y}.$$

[1 mark]

Module 3 formula glossary

PDE: Partial Differential Equation.

$p = \dfrac{\partial z}{\partial x}$ and $q = \dfrac{\partial z}{\partial y}$.

Lagrange linear PDE: $Pp + Qq = R$.

Auxiliary equations: $\dfrac{dx}{P} = \dfrac{dy}{Q} = \dfrac{dz}{R}$.

One-dimensional heat equation: $u_t = \alpha^2 u_{xx}$.

Thermal diffusivity: $\alpha^2 = K/(\rho c)$.

ODE: Ordinary Differential Equation.

$\cosh u = \dfrac{e^u + e^{-u}}{2}$.

Module 4 â€” Numerical Methods I

Q7

Q7(a) Apply four iterations of the Regula-Falsi method to approximate the positive root of

$$x^2 - 2 = 0$$

in $(1, 2)$.

(6 marks)

Intuition: Join the points at the ends of the sign-changing interval by a chord and use its x-intercept as the next approximation.

Solution and marking scheme

Let $f(x) = x^2 - 2$.

Since

$$f(1) = -1, \qquad f(2) = 2,$$

a root lies in $(1, 2)$. **[1 mark]**

Regula-Falsi formula:

$$x = \frac{af(b) - bf(a)}{f(b) - f(a)}.$$

[1 mark]

First iteration:

$$x_1 = \frac{1(2) - 2(-1)}{2 - (-1)} = \frac{4}{3} = 1.333333,$$

with $f(x_1) = -0.222222$. **[1 mark]**

Second iteration:

$$x_2 = \frac{7}{5} = 1.400000, \qquad f(x_2) = -0.04.$$

[1 mark]

Further iterations:

$$x_3 = \frac{24}{17} = 1.411765, \qquad x_4 = \frac{41}{29} = 1.413793.$$

[1 mark]

Therefore, after four iterations,

$$\boxed{x \approx 1.4138}.$$

The exact root is $\sqrt{2} \approx 1.4142$, confirming the iteration. **[1 mark]**

Q7(b) Using Newtonâ€™s forward interpolation formula, find y at $x = 1.5$:

x	0	1	2	3
y	1	2	5	10

(7 marks)

Intuition: The x-values are equally spaced, so construct forward differences from the beginning of the table.

Solution and marking scheme

Here,

$$x_0 = 0, \qquad h = 1, \qquad u = \frac{x - x_0}{h} = 1.5.$$

[1 mark]

Difference table:

y	Δy	$\Delta^2 y$	$\Delta^3 y$
1	1	2	0
2	3	2	
5	5		
10			

[2 marks]

Newtonâ€™s forward formula is

$$y = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \cdots.$$

[1 mark]

Substitute:

$$y = 1 + (1.5)(1) + \frac{(1.5)(0.5)}{2}(2).$$

[1 mark]

Thus,

$$y = 1 + 1.5 + 0.75.$$

[1 mark]

Therefore,

$$\boxed{y(1.5) = 3.25}.$$

[1 mark]

Q7© Use Simpsonâ€™s $1/3$ rule with six equal subintervals to evaluate

$$\int_0^1 \frac{dx}{1+x^2}. \qquad (7 \text{ marks})$$

Intuition: Simpsonâ€™s rule fits parabolas through consecutive groups of three ordinates.

Solution and marking scheme

Here,

$$h = \frac{1-0}{6} = \frac{1}{6}.$$

[1 mark]

The function values are approximately:

x	0	$1/6$	$2/6$	$3/6$	$4/6$	$5/6$	1
$f(x)$	1	0.972973	0.900000	0.800000	0.692308	0.590164	0.500000

[2 marks]

Simpsonâ€™s $1/3$ formula:

$$I \approx \frac{h}{3}[f_0 + f_6 + 4(f_1 + f_3 + f_5) + 2(f_2 + f_4)].$$

[1 mark]

Odd-index sum:

$$f_1 + f_3 + f_5 = 2.363137.$$

[1 mark]

Even-index sum:

$$f_2 + f_4 = 1.592308.$$

[1 mark]

Therefore,

$$I \approx \frac{1}{18}[1.5 + 4(2.363137) + 2(1.592308)] = \boxed{0.785398 \text{ approximately}}.$$

[1 mark]

The exact value is $\tan^{-1}(1) = \pi/4 \approx 0.785398$.

OR

Q8

Q8(a) Use the Newton-Raphson method to find $\sqrt{3}$, starting with $x_0 = 2$. Carry out three iterations.

(6 marks)

Intuition: Apply the tangent-intercept iteration to $f(x) = x^2 - 3$.

Solution and marking scheme

Take

$$f(x) = x^2 - 3, \qquad f'(x) = 2x.$$

[1 mark]

Newton-Raphson formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = \frac{1}{2}\left(x_n + \frac{3}{x_n}\right).$$

[1 mark]

First iteration:

$$x_1 = \frac{1}{2}\left(2 + \frac{3}{2}\right) = 1.75.$$

[1 mark]

Second iteration:

$$x_2 = \frac{1}{2}\left(1.75 + \frac{3}{1.75}\right) = 1.732142857.$$

[1 mark]

Third iteration:

$$x_3 = \frac{1}{2}\left(1.732142857 + \frac{3}{1.732142857}\right) = 1.732050810.$$

[1 mark]

Therefore,

$$\boxed{\sqrt{3} \approx 1.7321}.$$

[1 mark]

Q8 (b) Using Newtonâ€™s divided-difference formula, estimate $f(3)$:

x	1	2	4
$f(x)$	2	5	17

(7 marks)

Intuition: Divided differences work directly with unequal or equal intervals and build the interpolating polynomial incrementally.

Solution and marking scheme

First divided differences:

$$f[1,2] = \frac{5-2}{2-1} = 3.$$

[1 mark]

Also,

$$f[2,4] = \frac{17-5}{4-2} = 6.$$

[1 mark]

Second divided difference:

$$f[1,2,4] = \frac{6-3}{4-1} = 1.$$

[1 mark]

Newtonâ€™s polynomial is

$$P(x) = f(1) + (x-1)f[1,2] + (x-1)(x-2)f[1,2,4].$$

[1 mark]

Thus,

$$P(x) = 2 + 3(x-1) + (x-1)(x-2).$$

[1 mark]

At $x = 3$:

$$P(3) = 2 + 3(2) + (2)(1).$$

[1 mark]

Therefore,

$$\boxed{f(3) = 10}.$$

[1 mark]

Q8© Use Simpsonâ€™s $3/8$ rule with six equal subintervals to evaluate

$$\int_0^1 \frac{dx}{1+x}. \qquad \text{(7 marks)}$$

Intuition: The composite $3/8$ rule assigns weight 2 to interior indices divisible by 3 and weight 3 to the other interic indices.

Solution and marking scheme

Here,

$$n = 6, \qquad h = \frac{1}{6}.$$

[1 mark]

Values:

x	0	$1/6$	$2/6$	$3/6$	$4/6$	$5/6$	1
$f(x)$	1	0.857143	0.750000	0.666667	0.600000	0.545455	0.500000

[2 marks]

Composite Simpsonâ€™s $3/8$ formula:

$$I \approx \frac{3h}{8}[f_0 + f_6 + 3(f_1 + f_2 + f_4 + f_5) + 2f_3].$$

[1 mark]

The weight-3 sum is

$$f_1 + f_2 + f_4 + f_5 = 2.752598.$$

[1 mark]

Therefore,

$$I \approx \frac{1}{16}[1.5 + 3(2.752598) + 2(0.666667)].$$

[1 mark]

Hence,

$$\boxed{I \approx 0.693195}.$$

[1 mark]

The exact value is $\log 2 \approx 0.693147$.

Module 4 formula glossary

Regula-Falsi: Method of false position.

$$x = \frac{af(b) - bf(a)}{f(b) - f(a)}.$$

Newton–Raphson method:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Newton forward interpolation:

$$y = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \cdots.$$

Divided difference: Interpolation difference suitable for unequal spacing.

Simpsonâ€™s $1/3$ rule:

$$I \approx \frac{h}{3}[y_0 + y_n + 4\Sigma y_{\text{odd}} + 2\Sigma y_{\text{even}}].$$

Simpsonâ€™s $3/8$ rule: Requires the number of subintervals to be divisible by 3.

Module 5 â€” Numerical Methods II

Q9

Q9(a) Using Taylorâ€™s series up to the fourth-degree term, find $y(0.1)$ if

$$\frac{dy}{dx} = x + y, \qquad y(0) = 1. \qquad \text{(6 marks)}$$

Intuition: Obtain successive derivatives from the differential equation and substitute them into Taylorâ€™s expansion.

Solution and marking scheme

Taylorâ€™s formula:

$$y(x+h) = y + hy' + \frac{h^2}{2!}y'' + \frac{h^3}{3!}y''' + \frac{h^4}{4!}y''''.$$

[1 mark]

At $(0, 1)$:

$$y' = x + y = 1.$$

[1 mark]

Differentiate:

$$y'' = 1 + y' = 2.$$

[1 mark]

Further,

$$y''' = y'' = 2, \qquad y'''' = y''' = 2.$$

[1 mark]

With $h = 0.1$:

$$y(0.1) \approx 1 + 0.1 + \frac{2(0.1)^2}{2} + \frac{2(0.1)^3}{6} + \frac{2(0.1)^4}{24}.$$

[1 mark]

Therefore,

$$\boxed{y(0.1) \approx 1.110342}.$$

[1 mark]

Q9 (b) Using the fourth-order Runge-Kutta method, find $y(0.1)$ for

$$\frac{dy}{dx} = x + y, \qquad y(0) = 1,$$

taking $h = 0.1$.

(7 marks)

Intuition: RK4 estimates the weighted average slope using four slope calculations.

Solution and marking scheme

Using the increment convention $k_i = hf(\cdot)$:

Formula:

$$y_1 = y_0 + \frac{k_1 + 2k_2 + 2k_3 + k_4}{6}.$$

[1 mark]

First increment:

$$k_1 = 0.1f(0, 1) = 0.1.$$

[1 mark]

Second increment:

$$k_2 = 0.1f(0.05, 1.05) = 0.1(1.10) = 0.11.$$

[1 mark]

Third increment:

$$k_3 = 0.1f(0.05, 1.055) = 0.1(1.105) = 0.1105.$$

[1 mark]

Fourth increment:

$$k_4 = 0.1f(0.1, 1.1105) = 0.1(1.2105) = 0.12105.$$

[1 mark]

Hence,

$$y_1 = 1 + \frac{0.1 + 2(0.11) + 2(0.1105) + 0.12105}{6}.$$

[1 mark]

Therefore,

$$\boxed{y(0.1) \approx 1.110342}.$$

[1 mark]

Q9© Given

$$\frac{dy}{dx} = x$$

and

$$y(0) = 1, \quad y(0.1) = 1.005, \quad y(0.2) = 1.020, \quad y(0.3) = 1.045,$$

use Milneâ€™s predictor-corrector method to find $y(0.4)$.

(7 marks)

Intuition: Use four known values to predict the next value and then improve it with the corrector formula.

Solution and marking scheme

Here,

$$h = 0.1, \qquad f_i = x_i.$$

Thus,

$$f_0 = 0,\ f_1 = 0.1,\ f_2 = 0.2,\ f_3 = 0.3.$$

[1 mark]

Milneâ€™s predictor:

$$y_4^{(p)} = y_0 + \frac{4h}{3}(2f_1 - f_2 + 2f_3).$$

[1 mark]

Substitute:

$$y_4^{(p)} = 1 + \frac{0.4}{3}[2(0.1) - 0.2 + 2(0.3)].$$

[1 mark]

Hence,

$$y_4^{(p)} = 1 + \frac{0.4}{3}(0.6) = 1.08.$$

[1 mark]

At $x_4 = 0.4$:

$$f_4^{(p)} = 0.4.$$

[1 mark]

Milneâ€™s corrector:

$$y_4^{(c)} = y_2 + \frac{h}{3}(f_2 + 4f_3 + f_4^{(p)}).$$

Thus,

$$y_4^{(c)} = 1.02 + \frac{0.1}{3}(0.2 + 1.2 + 0.4).$$

[1 mark]

Therefore,

$$\boxed{y(0.4) = 1.080}.$$

[1 mark]

OR

Q10

Q10 (a) Use the modified Euler method to find $y(0.1)$ for

$$\frac{dy}{dx} = x + y, \qquad y(0) = 1,$$

taking $h = 0.1$ and three corrections.

(6 marks)

Intuition: First predict using the starting slope, then repeatedly average the starting and ending slopes.

Solution and marking scheme

Predictor:

$$y_1^{(0)} = y_0 + hf(x_0, y_0) = 1 + 0.1(1) = 1.1.$$

[1 mark]

Corrector formula:

$$y_1^{(r+1)} = y_0 + \frac{h}{2}[f(x_0, y_0) + f(x_1, y_1^{(r)})].$$

[1 mark]

First correction:

$$y_1^{(1)} = 1 + 0.05[1 + (0.1 + 1.1)] = 1.11.$$

[1 mark]

Second correction:

$$y_1^{(2)} = 1 + 0.05[1 + (0.1 + 1.11)] = 1.1105.$$

[1 mark]

Third correction:

$$y_1^{(3)} = 1 + 0.05[1 + (0.1 + 1.1105)] = 1.110525.$$

[1 mark]

Therefore,

$$\boxed{y(0.1) \approx 1.1105}.$$

[1 mark]

Q10 (b) Use the fourth-order Runge-Kutta method to find $y(0.2)$ for

$$\frac{dy}{dx} = 2x + y, \qquad y(0) = 1,$$

taking $h = 0.2$.

(7 marks)

Intuition: Calculate four representative slope increments over the interval and combine them using RK4 weights.

Solution and marking scheme

Use

$$y_1 = y_0 + \frac{k_1 + 2k_2 + 2k_3 + k_4}{6}.$$

[1 mark]

First increment:

$$k_1 = 0.2f(0, 1) = 0.2.$$

[1 mark]

Second increment:

$$k_2 = 0.2f(0.1, 1.1) = 0.2(0.2 + 1.1) = 0.26.$$

[1 mark]

Third increment:

$$k_3 = 0.2f(0.1, 1.13) = 0.2(0.2 + 1.13) = 0.266.$$

[1 mark]

Fourth increment:

$$k_4 = 0.2f(0.2, 1.266) = 0.2(0.4 + 1.266) = 0.3332.$$

[1 mark]

Therefore,

$$y_1 = 1 + \frac{0.2 + 2(0.26) + 2(0.266) + 0.3332}{6}.$$

[1 mark]

Hence,

$$\boxed{y(0.2) = 1.2642}.$$

[1 mark]

Q10© Given

$$\frac{dy}{dx} = x + y$$

and

x	0	0.1	0.2	0.3
y	1.000000	1.110340	1.242806	1.399718

use Milneâ€™s predictor-corrector method to find $y(0.4)$. Use two corrections.

(7 marks)

Intuition: Compute the derivative at each known point, predict y_4, and then apply the corrector twice.

Solution and marking scheme

Since $f = x + y$:

$$f_0 = 1, \quad f_1 = 1.210340, \quad f_2 = 1.442806, \quad f_3 = 1.699718.$$

[1 mark]

Predictor:

$$y_4^{(p)} = y_0 + \frac{4h}{3}(2f_1 - f_2 + 2f_3).$$

[1 mark]

Hence,

$$y_4^{(p)} = 1 + \frac{0.4}{3}[2(1.210340) - 1.442806 + 2(1.699718)] = 1.583641.$$

[1 mark]

Therefore,

$$f_4^{(p)} = 0.4 + 1.583641 = 1.983641.$$

[1 mark]

First correction:

$$y_4^{(1)} = y_2 + \frac{h}{3}(f_2 + 4f_3 + f_4^{(p)}) = 1.583649.$$

[1 mark]

Update

$$f_4^{(1)} = 0.4 + 1.583649 = 1.983649,$$

and correct again:

$$y_4^{(2)} = 1.242806 + \frac{0.1}{3}[1.442806 + 4(1.699718) + 1.983649] = 1.583650.$$

[1 mark]

Therefore,

$$\boxed{y(0.4) \approx 1.58365}.$$

[1 mark]

Module 5 formula glossary

ODE: Ordinary Differential Equation.

Taylor method:

$$y(x+h) = y + hy' + \frac{h^2}{2!}y'' + \cdots.$$

Modified Euler method: Predictor followed by trapezoidal-slope corrections.

RK4: Fourth-order Runge-Kutta method.

Milne predictor:

$$y_{n+1}^{(p)} = y_{n-3} + \frac{4h}{3}(2f_{n-2} - f_{n-1} + 2f_n).$$

Milne corrector:

$$y_{n+1}^{(c)} = y_{n-1} + \frac{h}{3}(f_{n-1} + 4f_n + f_{n+1}).$$

In numerical ODE formulas, $f_i = f(x_i, y_i)$.

Arithmetic-verification summary

Every full question totals $6+7+7=20$ **marks**.

Module 1 exact results were independently checked by direct integration.

The Green and Stokes results agree with direct geometric interpretations.

PDE solutions satisfy the original differential equations and supplied conditions.

Simpsonâ€™s approximations agree with:

$$\frac{\pi}{4} \approx 0.785398, \qquad \log 2 \approx 0.693147.$$

Taylor and RK4 for $y' = x + y$, $y(0) = 1$ both produce approximately

$$y(0.1) = 1.110342.$$

The exact value for the final Milne problem is approximately 1.583649, consistent with the computed result.